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Unit 15: Project Management KPIs

BSS 310 Engineering Management 2026

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Quick Recap/Questions arising from Previous Lecture

The Lecture addressed:

- Business systems intelligence spanning across “behavioural” and “sentimental” intelligence respectively addressed using “systems thinking” and “Naïve Bayes theorem”
- Risk identification, analysis and evaluation with the aid of developed tables of metrication and the risk chart.

Learning Outcomes for the Current Lecture

At the end of this lecture you should be able to:

- **Define**, **discuss**, **apply**, and **evaluate** facts about KPIs, including distinguishing between KPIs, PIs, and/or Metrics.

- **Discuss** and **analyse** the Anatomy of a KPI, Attributes of a Good KPI; **Discuss** and **analyse** KPIs often used in Project Management, and **formulate** KPIs for the semester project.

Presentation Outline

- Definition of Key performance indicators (KPIs)
- Some facts about KPIs
- Anatomy of a KPI
- Attributes of a Good KPI
- Generic KPIs for Project Management
- KPIs formulation for the semester project

Definition of KPIs

Key performance indicators (KPIs) in project management is defined as a set of critically important project variables that must be identified, measured, monitored and managed in a bid to gain insight into the performance of a project.

Project management KPIs are generally agreed upon early in the life of a project. They reflect an organisation’s central concept of a project and solidify project responsibility across administrative divisions.

Project management Dashboard

Some Facts about KPIs (Metrics and/or PIs)

- The terms “metrics” or “PIs” are generic whereas “KPIs” are specific. KPIs serve as early warning signs that, if an unfavourable condition exists and is not addressed, the results could be poor. KPIs and metrics can be displayed in dashboards, scorecards, and reports.

- Part of the project manager’s role is to understand what the critical metrics or variables are that need to be identified, measured, reported, and managed such that the project will be viewed as a success by all the stakeholders.

Some Facts about KPIs (Who identifies or creates a KPI?)

- **Defining the correct metrics or KPIs is a joint venture between the project manager and stakeholders.** This is necessary to secure stakeholder agreement.

- **One of the keys to a successful project is the effective and timely management of information.** KPIs provide information to make informed decisions by reducing uncertainty.

Some Facts about KPIs (Considering Duration)

Unlike **long-term business environments**, **project environments** are much shorter and therefore more susceptible to changing metrics. In a project environment, metrics can change from project to project, during each life-cycle phase, and at any time.

Some Facts that can Impact on KPI formation

- The way the company defines value internally
- The way the customer and contractor agree to project initiation as to what metrics should be used on a given project
- New or updated versions of tracking software
- Improvements to the enterprise project management methodology and accompanying project management information system
- Changes in the enterprise environmental factors
- Changes in the project's business case assumptions

Anatomy of a KPI (Analysing the term KPI)

Some metrics, such as project profitability, can tell us if things look good or bad but do not necessarily provide meaningful information on what we must do to improve performance. Therefore, a typical KPI must do more than just function as a metric. If we dissect the KPIs, we will see the following:

- KEY = a major contributor to the success or failure of the project. A KPI metric is therefore only “key” when it can make or break the project.
- PERFORMANCE = a form of appraisal; a metric that can be measured, relied upon for decision making.
- INDICATOR = a metric that can be viewed, displayed, and reasonably provide quantifications for comparing the present and future performances.

Anatomy of a KPI (Exploring the CSFs)

Defining and selecting the KPIs are much easier if you define the Critical Success Factor (CSFs) first. KPIs should not be confused with CSFs. CSFs are things that must be in place to achieve the KPIs or system objectives. A KPI is not a CSF but may provide a leading indication that the CSF can be met.

Selecting the right KPIs and the right number of KPIs will:

- Allow for better decision making
- Improve performance on the project
- Help identify problem areas faster
- Improve customer–contractor–stakeholder relations

Anatomy of a KPI

David Parameter Defined Three Categories of Metrics:

- Results Indicators (RIs): What have we accomplished?
- Performance Indicators (PIs): What must we do to increase or meet performance?
- Key Performance Indicators (KPIs): What are the Critical Systemic Variables that can drastically increase performance or accomplishment of the objectives?

The number of KPIs can vary from project to project and may be impacted by the number of stakeholders. Some people select the number of KPIs based on the Pareto principle, which states that 20 percent of the total indicators will impact 80 percent of the project outcome.

Anatomy of a KPI

David Parameter states that the 10/80/10 rule is usually applied when selecting the number of KPIs:

- RIs: 10
- PIs: 80
- KPIs: 10

Typically, between six and ten KPIs are standard. Factors influencing the number of KPIs include:

- The number of information systems that the project manager uses (i.e., one, two, or three, etc.)
- The number of stakeholders and their reporting requirements
- The ability to measure the information

Anatomy of a KPI

- The organizational process assets available to collect information
- The cost of measurement and collection
- Dashboard reporting limitations

Attributes of a Good KPI

The literature abounds with articles defining the attributes of metrics and KPIs. All too often, authors use the "SMART" rule as a means of identifying the attributes:

- S = Specific: clear and focused toward performance targets or a business purpose
- M = Measurable: can be expressed quantitatively
- A = Attainable: the targets are achievable
- R = Realistic or relevant: the KPI is reasonable or directly pertinent to the work done on the project
- T = Time-based: the KPI is measurable within a given time period

Attributes of a Good KPI

The SMART rule was originally developed for establishing meaningful objectives for projects and later adapted to the identification of metrics and KPIs. While the use of the SMART rule does have some merit, its applicability to KPIs is occasionally questionable.

KPIs for Project Management

Deviation of Planned Hours of Work

- Understanding what tasks took more or less time can help in the efficient allocation of consulting and training time. In addition, understanding which teams had to go above and beyond can help build a meaningful incentive and reward program, as well as improve time allocation planning.

KPIs for Project Management

Percentage of Milestones Missed

- Here, a project completion time due date is seen to be shifting over its lifecycle. Identifying milestones and achieving goals are important for maintaining project momentum. When too many milestones are not achieved or are shifted, employees may feel frustrated. Identifying when milestones are missed can help restart a project and mitigate similar challenges in the future.

KPIs for Project Management

Planned Value (PV)

- This metric is also referred to as Budgeted Cost of Work Scheduled (BCWS). The planned value is the estimated cost for the project activities planned/scheduled as of reporting date. Compare the Planned Value with the other project KPIs to see whether you're running ahead of schedule or have already spent a bigger slice of your budget than scheduled to date. Knowing how, where and why project budget deviated is important in tracking down waste and inefficiency, as well as for planning better for the unforeseen challenges inherent to most projects.

KPIs for Project Management

Actual Cost (AC)

- The Actual Cost KPI is also referred to as the Actual Cost of Work Performed (ACWP). It indicates how much money you have spent on a project to date.

KPIs for Project Management

Effective Cost Variance Documentation

- Keeping accurate records related to cost variance can provide a detailed profile of which teams and processes are most efficient. It can also help companies decide whether a project was worth the investment and assist managers in deciding whether to initiate similar projects.

KPIs for Project Management

Cost of managing processes

- Add this metric to your project dashboard to get an overview of the time and resources spent on supervising and managing the project. If the cost of managing processes seems too high, it might indicate that your project manager is doing an inefficient job. On the other hand, if the management costs are too low, it means that your teamwork may be poorly organised. It’s normal for the team to spend time on project meetings, making sure that everyone is on the same page.

KPIs for Project Management

Return on Investment (ROI)

- Project's ROI reflects on its profitability and shows whether the benefits of the project exceed its cost. Not all projects are destined to have a positive ROI in the first place. Sometimes, the ROI should be considered long-term as some projects take more time to grow profits. ROI metrics on the project KPI dashboard should be derived from measurable components like the project's Actual Cost and Earned Value.

KPIs for Project Management

Overdue project tasks / crossed deadlines

- Add this metric to your project tracking dashboard to get an overview of how many project activities are overdue. This KPI is a calculated percentage of projects with crossed deadlines compared to all the completed project activities. If you have a high percentage of overdue tasks, it's time to think through the project schedule and bring in some new contributors.

KPIs for Project Management

Percentage of tasks completed

- To get a really quick overview of the project's performance, create a KPI indicating the percentage of completed tasks. This KPI does not guarantee efficiency, but rather the effectiveness of the project team in respect of picking up a task and being sure of completing it, however, without any time affiliation.

KPIs for Project Management

Percentage of projects completed on time

- If you’re frequently handling multiple projects, this KPI is a must-have on your project management dashboard. This project KPI indicates the number of projects completed on time compared to crossed deadlines. If you’re not able to keep this percentage over 80%, it might be time to hire some new team members or accept fewer projects from customers.

KPIs for Project Management

Percentage of cancelled projects

- Similarly to the previous project performance indicator, this indicator is widely used by agencies that frequently take up new projects. But the number of cancelled projects also reflects on a company's capability to plan ahead. Foreseeing whether a project is going to be profitable and whether the team is sufficiently qualified to achieve all the project goals. Perceive the percentage of cancelled projects as a reflection on the sustainability of your business decisions.

KPIs for the Semester Project

Business Related KPIs

- Sales volume
- Production volume
- Customer satisfaction
- Staff competence
- Break-even point and time
- Profit margin etc

Lecture Resources

Recommended reading in the handbook:

- Project Management - A Multi-disciplinary Approach (4th edition) Chapter 15: Metrics

Others

- floridatechonline.com/blog/business/key-performance-indicators-in-project-management/#:~:text=Top%20project%20management%20benchmarking%20measures,KPIs%20within%20project%20management%20include%3A&text=Labor%20costs%20spent%20per%20month

  https://www.scoro.com/blog/16-essential-project-kpis/

Learning Outcomes Evaluated

- Are you able to define, discuss, analyse, and evaluate a KPI?
- Can you distinguish between a KPI, PI, and/or a Metric?
- Can you discuss and analyse the Anatomy of a KPI?
- Do you know some attributes of a good KPI?
- Can you identify, discuss, and analyse KPIs often used in Project Management?
- Can you formulate KPIs for your business in the semester project?

What is your standing with respect to the above queries?

Next Topic

- Unit 16: Organisational Structure

The End